🦞 OpenClaw: Complete Guide 0 → 100%

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> ***Creation Date:*** *2026-04-02*
> ***Inspired by:*** *openclaw-ops by @cathrynlavery*

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1. Initial Installation

Prerequisites

Tool	Version	Command
Node.js	22.16+ or 24	`node --version`
npm/pnpm/bun	Latest	`npm --version`
Git	Latest	`git --version`
Python 3	3.9+	`python3 --version`

Installation

```
# Install OpenClaw globally
npm install -g openclaw@latest

# Or with pnpm (recommended)
pnpm add -g openclaw@latest

# Verify installation
openclaw --version
```

Onboarding

```
# Interactive setup assistant
openclaw onboard --install-daemon

# This configures:
# - Gateway daemon
# - Initial workspace
# - Environment variables
# - Auto-start daemon
```

Initial Verification

```
# Gateway status
openclaw gateway status

# General status
openclaw status

# View logs in real-time
tail -f ~/.openclaw/logs/gateway.log
```

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2. Basic Configuration

Directory Structure

```
~/.openclaw/
├── openclaw.json          # Main configuration
```

```
├── workspace/              # Your workspace
│   ├── SOUL.md            # Agent personality
│   ├── AGENTS.md          # Workspace conventions
│   ├── MEMORY.md          # Long-term memory
│   ├── memory/            # Daily notes
│   ├── tasks/             # Task management
│   └── skills/            # Custom skills
├── skills/                 # Installed skills
└── logs/                   # System logs
```

Minimum Configuration (openclaw.json)

```
{
  "agent": {
    "model": "modelstudio/qwen3.5-plus"
  },
  "providers": {
    "nvidia": {
      "apiKey": "nvapi-xxxxx",
      "baseUrl": "https://integrate.api.nvidia.com/v1"
    },
    "modelstudio": {
      "apiKey": "sk-xxxxx",
      "baseUrl": "https://dashscope.aliyuncs.com/compatible-mode/v1"
    }
  },
  "env": {
    "EXA_API_KEY": "xxxxx"
  },
  "security": {
    "actionApproval": {
      "required": ["email.send", "file.delete", "shell.exec"]
    }
  }
}
```

Environment Variables (.zshenv or .bashrc)

```
# NVIDIA NIM (FREE tier)
export NVIDIA_API_KEY="nvapi-xxxxx"
```

```
# Alibaba DashScope
export DASHSCOPE_API_KEY="sk-xxxxx"

# MiniMax
export MINIMAX_API_KEY="xxxxx"

# Exa AI (web searches)
export EXA_API_KEY="xxxxx"
```

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3. Models and APIs

Recommended Providers

Provider	Model	Context	Cost	Use
NVIDIA NIM	minimax-m2.5	32K	FREE	Daily
Alibaba	qwen3.5-plus	256K	~$0.10/1M	Production
Alibaba	qwen-plus	128K-1M	~$0.40/1M	Long docs
Alibaba	qwen-turbo	128K	~$0.05/1M	Fast
Local	qwen3.5:9b	32K	FREE	Offline

Fallback Chain

```
MODELS = [
    "minimaxai/minimax-m2.5",            // PRIMARY
    "nvidia/nemotron-4-340b-instruct",   // Fallback 1
    "deepseek-ai/deepseek-r1-distill-qwen-32b",  // Fallback 2
    "meta/llama-3.1-70b-instruct",       // Fallback 3
]
```

Estimated Monthly Cost (100K tokens/day)

Provider	Cost/Mo
NVIDIA NIM	$0.00
Alibaba Qwen	~$0.50
TOTAL	**~$0.50**

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4. Communication Channels

Supported Channels

- ✅ Telegram
- ✅ Discord
- ✅ WhatsApp
- ✅ Slack
- ✅ Signal
- ✅ iMessage (macOS)
- ✅ Google Chat
- ✅ IRC
- ✅ Matrix
- ✅ LINE

Telegram Configuration (Example)

```
{
  "channels": {
    "telegram": {
      "botToken": "xxxxx:xxxxx",
      "allowedChats": ["-1003742752506"]
    }
  }
}
```

Channel Diagnostics

```
# List channels
openclaw channels list

# Test connection
openclaw channels test telegram
```

Common Issues

Channel	Issue	Solution
Slack	`missing_scope`	Renew token with correct scopes

Channel	Issue	Solution
WhatsApp	Disconnection	Use Bun instead of Node
Telegram	Not responding	Verify bot token
iMessage	No access	Enable Full Disk Access
Discord	WebSocket 1005/1006	Restart gateway

5. Essential Skills

Skill Installation

```
# Official skills
openclaw skills install weather
openclaw skills install github
openclaw skills install summarize

# Third-party skills (audit first!)
git clone https://github.com/cathrynlavery/openclaw-ops.git
        ~/.openclaw/skills/openclaw-ops
```

Recommended Skills

Skill	Purpose	Priority
`openclaw-ops`	Monitoring & health	**Critical**
`skill-vetter`	Skill security audit	**Critical**
`weather`	Weather & forecasts	High
`github`	Issues/PRs management	High
`summarize`	Summarize URLs/ transcripts	High
`healthcheck`	Security hardening	High
`model-usage`	Cost tracking	Medium
`memory`	Memory management	Medium

Skill Audit (MANDATORY)

```
# Before installing any third-party skill
cd ~/.openclaw/skills/openclaw-ops
bash scripts/skill-audit.sh <skill-path>

# Check for red flags:
# - Hardcoded secrets
# - Suspicious network calls
# - Prompt injection patterns
```

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6. Operations and Monitoring

openclaw-ops: Essential Scripts

Script	Purpose	Frequency
`heal.sh`	Auto-repair	On-demand
`watchdog.sh`	Continuous monitoring	Every 5 min
`check-update.sh`	Post-update triage	Post-update
`security-scan.sh`	Security audit	Weekly
`health-check.sh`	Health checks	Hourly

Watchdog Installation (macOS)

```
cd ~/.openclaw/skills/openclaw-ops
bash scripts/watchdog-install.sh

# View logs
tail -f ~/.openclaw/logs/watchdog.log

# View incidents
cat ~/.openclaw/logs/heal-incidents.jsonl
```

Watchdog on Linux (cron)

```
# Edit crontab
crontab -e

# Add every 5 minutes
*/5 * * * * bash /path/to/scripts/watchdog.sh >> ~/.openclaw/
        logs/watchdog.log 2>&1
```

Health Targets (health-targets.conf)

```
# Copy template
cp templates/health-targets.conf.example ~/.openclaw/health-
        targets.conf

# Example configuration
gateway http://localhost:18789/health 300
knowledge http://localhost:8001/health 300
ops http://localhost:8002/health 300
```

Run Health Check

```
cd ~/.openclaw/skills/openclaw-ops
bash scripts/health-check.sh --verbose
```

---

7. Security and Hardening

Security Scan

```
cd ~/.openclaw/skills/openclaw-ops
bash scripts/security-scan.sh

# Generates report with:
# - Hardening score (0-100)
# - Exposed credentials
# - File permissions
# - Insecure configuration
```

Hardening Checklist

- [ ] `auth: "none"` removed (v2026.1.29+)
- [ ] API keys in environment variables, NOT in config
- [ ] `tools.exec.ask` and `tools.exec.security` configured
- [ ] Third-party skills audited before install
- [ ] Correct file permissions (600 for secrets)
- [ ] No secrets in logs or history
- [ ] Firewall configured for gateway
- [ ] Tailscale/SSH tunneled for remote access

Critical CVEs (v2026.2.12+)

CVE	Description	Impact
CVE-2026-25253	RCE via token leakage	**Critical**
SSRF-2026-001	SSRF in MCP tools	High
PATH-2026-002	Path traversal	High
PROMPT-2026-003	Prompt injection	Medium

⚠️ **Minimum version: v2026.2.12**

File Permissions

```
# Config with secrets
chmod 600 ~/.openclaw/openclaw.json

# Executable scripts
chmod 755 ~/.openclaw/skills/*/scripts/*.sh

# Logs (read-only for user)
chmod 644 ~/.openclaw/logs/*.log
```

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8. Maintenance

Daily (Automatic via Heartbeat)

- [ ] Check urgent emails
- [ ] Check calendar (next 24-48h)
- [ ] Check notifications
- [ ] Rotate logs if >100MB

Weekly

```
# Security scan
cd ~/.openclaw/skills/openclaw-ops
bash scripts/security-scan.sh

# Skill audit (if installed new skills)
bash scripts/skill-audit.sh ~/.openclaw/skills/<new-skill>

# Clean sessions
openclaw sessions prune --older-than 7d
```

Monthly

- [ ] Review MEMORY.md and curate insights
- [ ] Update openclaw: `openclaw update`
- [ ] Run `check-update.sh` post-update
- [ ] Review API costs
- [ ] Backup workspace and config

Post-Update Triage

```
cd ~/.openclaw/skills/openclaw-ops

# Report only
bash scripts/check-update.sh

# Report + auto-fix
bash scripts/check-update.sh --fix
```

Common Post-Update Issues

Issue	Cause	Solution
Gateway won't start	`auth: "none"` removed	Configure valid auth
Agents blocked	Exec approvals reset	Re-configure allowlists
Channels offline	Expired tokens	Renew tokens
Skills failing	Breaking changes	Review changelog

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9. Troubleshooting

Gateway Down

```
# Check status
openclaw gateway status

# View error logs
tail -100 ~/.openclaw/logs/gateway.err.log

# Try restart
openclaw gateway restart

# If fails, run heal
cd ~/.openclaw/skills/openclaw-ops
bash scripts/heal.sh
```

Agents Stuck on /approve

```
# Check exec config
openclaw config get tools.exec

# Common fix: empty allowlist shadowing wildcard
# Edit ~/.openclaw/openclaw.json and remove empty entries
```

Channels Not Responding

```
# List channels
openclaw channels list
```

```
# Test individual
openclaw channels test telegram

# View channel logs
tail -f ~/.openclaw/logs/channels.log
```

Memory Full / Context Overflow

```
# Check context usage
openclaw status

# Compact conversation
openclaw sessions compact

# Prune old sessions
openclaw sessions prune --older-than 7d
```

Diagnostic Commands

```
# Full status
openclaw status

# Version
openclaw --version

# Doctor (automatic diagnosis)
openclaw doctor

# List active sessions
openclaw sessions list

# View logs live
tail -f ~/.openclaw/logs/gateway.log
```

---

10. Final Checklist

Installation ✅

- [ ] Node.js 22.16+ or 24 installed
- [ ] OpenClaw installed globally
- [ ] Gateway daemon running
- [ ] Workspace initialized

Configuration ✅

- [ ] openclaw.json configured
- [ ] Environment variables in .zshenv/.bashrc
- [ ] Models configured with fallback
- [ ] At least 1 channel connected and tested

Skills ✅

- [ ] openclaw-ops installed
- [ ] skill-vetter installed
- [ ] watchdog installed (LaunchAgent or cron)
- [ ] Basic skills installed (weather, github, summarize)

Security ✅

- [ ] Security scan executed (score >80)
- [ ] API keys in env vars, not in config
- [ ] Correct file permissions
- [ ] Third-party skills audited

Monitoring ✅

- [ ] Watchdog running
- [ ] Health targets configured
- [ ] Logs being rotated
- [ ] Alerts configured (Signal/Telegram)

Maintenance ✅

- [ ] Heartbeat configured
- [ ] Automatic backup configured
- [ ] Documentation updated
- [ ] MEMORY.md being curated

📞 Support and Resources

Resource	URL
Official Docs	https://docs.openclaw.ai
GitHub	https://github.com/openclaw/openclaw
Discord	https://discord.com/invite/clawd
ClawHub (skills)	https://clawhub.ai
openclaw-ops	https://github.com/cathrynlavery/openclaw-ops

🎯 Next Steps

1. **Day 1:** Installation + basic config + 1 channel
2. **Day 2:** Install openclaw-ops + watchdog
3. **Day 3:** Security scan + hardening
4. **Day 4:** Configure heartbeat + automations
5. **Day 5:** Full testing + documentation

Guide created for OpenClaw production setup. Update according to new versions.

👤 Authorship

Role	Responsible
Created by	Friday Assistant
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Date	2026-04-02

Role	Responsible
Group	AI multiverse
Inspired by	openclaw-ops by @cathrynlavery

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